

Cross-diagnostic consensus for tokamak disruption prediction

A ten-iteration honest-iteration log from address-space novelty catcher to gradient-boosted multi-scale classifier on the 500-shot MAST dataset, with stability cross-validation and failure analysis

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Abstract

We document a ten-iteration honest-engineering record of porting a cross-diagnostic consensus catcher — originally developed for unmodeled gravitational-wave burst search and for cross-observable geometric-locus identification in spacetime metrics — to the problem of tokamak disruption prediction on the public 500-shot MAST dataset (Sharma 2025, Zenodo 10.5281/zenodo.16032053). The pipeline operates on the same robust rolling z-score + multi-detector consensus structure used by the Systrophē v2 detector triple: per-channel novelty scoring with median-and-MAD baseline, followed by an alert rule that requires a fraction-threshold of channels to co-fire simultaneously for a minimum sustain. We iterate the pipeline through nine design variations, document two outright negative results (hybrid pendulum/Kuramoto/Hamiltonian voting; failure-analysis-informed lean feature set), and converge on a 27-feature gradient-boosting classifier that achieves **$74.3 \pm 1.0\%$ TPR at $14.1 \pm 2.3\%$ FAR with 80.8 ± 2.4 ms median time-to-disruption** (mean $\pm$ std across three 5-fold cross-validation seeds). The result matches published recurrent-neural-network baselines on FAR while remaining ~ 11 pp below them on TPR. We give an explicit failure analysis showing that only 2% of disruptions are structurally invisible to our feature set, identify the remaining 24% miss budget as a threshold-tuning gap rather than a missing-information gap, and discuss why the remaining performance gap to RNN baselines likely requires per-shot operational metadata (disruption-cause classification) that this dataset does not expose. All code, all checkpoints, and all per-iteration JSON results are publicly available; the iteration log is reproducible commit-by-commit.

1 Introduction

A tokamak *disruption* is the sudden uncontrolled termination of a magnetic confinement plasma: stored magnetic energy that took seconds to build up is released within ~ 10 ms, depositing localized megajoule heat loads onto the wall and inducing eddy-current forces on the vessel structure. On ITER and on future fusion reactors, the engineering cost of an unmitigated disruption is sufficient that operations cannot proceed without a reliable forward-prediction system that triggers mitigation hardware (massive-gas-injection valves, shattered-pellet injectors) with adequate lead time, typically 30–50 ms [?, ?]. The standard performance figure is true-positive rate (TPR, fraction of real disruptions caught at ≥ 30 ms lead) versus false-alarm rate (FAR, fraction of non-disruptive shots triggering a spurious alert).

The published state of the art on real plasma data is recurrent-neural-network ensembles. Ferreira et al. (2019) [?] report a Long Short-Term Memory architecture on JET bolometer data

yielding approximately 85% TPR at 15% FAR. The TPR/FAR trade-off is a free design parameter (set by the binary threshold on the LSTM output) and the operating curve is dataset-dependent.

We are interested in a methodologically distinct approach: applying the *cross-observable consensus catcher* originally developed for unmodeled-burst search in gravitational-wave data [?] and for cross-observable identification of metric special loci in spacetime physics [?] to the tokamak disruption problem. The structural argument is that a disruption is a regime in which many independent plasma observables (magnetic, kinetic, radiative) deflect from their baseline simultaneously; this is exactly the cross-observable consensus signature the catcher is designed to detect. The empirical question is whether the methodology, applied without machine-specific tuning or operational metadata, can reach performance comparable to published RNN baselines.

This paper records a ten-iteration log of that effort. Of the ten iterations, eight produced measurable forward progress and two produced informative negative results that constrained subsequent feature design. The final operating point matches published RNN FAR at ~ 11 pp lower TPR; we identify the remaining gap explicitly.

2 Method

2.1 The catcher v2 detector triple

Given a C -channel diagnostic time series sampled at rate f_s , the catcher v2 operates as follows.

Per-channel novelty score. For each channel c and each sample t , the catcher computes a robust z-score of the assessment window against a trailing baseline:

$$z_c(t) = \frac{\text{median}(x_c[t - A : t]) - \text{median}(x_c[t - B : t - A])}{1.4826 \cdot \text{MAD}(x_c[t - B : t - A]) + \epsilon}$$

with assessment-window length A (we use 5 ms) and baseline-window length B (we use 50, 200, and 500 ms at three scales). A channel is flagged as novel at t if $|z_c(t)| > z_{\text{thr}}$ (we use 3.0). The median-and-MAD form is robust to channel-specific outlier distributions and requires no prior calibration.

Consensus alert. At each t , the catcher computes the fraction of channels novel:

$$f(t) = \frac{1}{C} \sum_{c=1}^C \mathbb{I}[|z_c(t)| > z_{\text{thr}}].$$

An alert fires when $f(t) \geq f_{\text{thr}}$ sustained for at least τ_{min} samples. The fraction threshold f_{thr} is the principal tunable parameter.

2.2 Per-channel weighting

Vanilla consensus weights every channel equally. We learn per-channel log-likelihood-ratio weights from labelled training shots:

$$w_c = \log \frac{P(|z_c| > z_{\text{thr}} \mid \text{precursor}) + \alpha}{P(|z_c| > z_{\text{thr}} \mid \text{quiescent}) + \alpha},$$

with Laplace smoothing $\alpha = 1$ over the binary-event counts. We clamp $w_c \geq 0$: a channel that fires more in quiescence than in precursor adds noise rather than signal. The weighted consensus then replaces $f(t)$:

$$f_w(t) = \frac{\sum_c w_c \mathbb{I}[|z_c(t)| > z_{\text{thr}}]}{\sum_c w_c}.$$

The weights are learned on training-fold shots only; the test fold sees only the inferred weights.

2.3 Multi-scale features

A single baseline window cannot capture both fast (locked-mode MHD) and slow (radiation collapse, density limit) precursors. We compute the per-channel z-score at three scales $B \in \{50, 200, 500\}$ ms, then derive the following per-time-step features:

$$\{f_w(t), R(t), R_{\text{excess}}(t), n_{\text{novel}}(t)\} \times 3 \text{ scales},$$

plus two time-derivatives ($\dot{f}_w$ and $\dot{R}_{\text{excess}}$ at the 200 ms scale), for 14 features. Here $R(t)$ is the Kuramoto order parameter computed from the analytic phases of the channel z-scores (Hilbert transform), and R_{excess} is $R(t)$ minus its trailing rolling median (window 300–600 samples depending on the scale).

2.4 Per-channel peak features

We add ten further features: $\max |z_c|$ within a trailing 500 ms window per channel. This gives the classifier discrimination by *which* channels are most active, not just how many channels co-fire.

2.5 Pulse-phase indicators

Three binary indicators encode the pulse phase from the ramp-up/flat-top/ramp-down boundaries in the dataset metadata. These let the classifier learn phase-conditional thresholds (e.g. “ignore Kuramoto rise during controlled ramp-down”). Crucially, the training sampler must include non-disruptive ramp-down windows as negatives to avoid the phase indicator becoming a label proxy; we document this explicit fix in §4.5.

2.6 Classifier

We use the scikit-learn `GradientBoostingClassifier` with 250 trees, max-depth 5, learning rate 0.05, subsample 0.7, minimum samples per leaf 10. Class imbalance ($\sim 71/429$ in our windowed sample after dense extraction) is handled by per-sample weights inversely proportional to class frequency. The classifier’s per-window posterior $P(t)$ feeds into a final alert rule: $P(t) \geq p_{\text{thr}}$ sustained for τ_{min} samples.

2.7 Alert masking

To avoid alerts firing on plasma ramp-up artefacts or on legitimate controlled ramp-downs, we restrict the alert evaluation window to

$$[t_{\text{flat_top_min}} + 50 \text{ ms}, \min(t_{\text{disrupt}}, t_{\text{flat_top_max}} + 20 \text{ ms})].$$

This baseline-shifted lower bound ensures the rolling baseline window has filled with flat-top data before any novelty assessment.

3 Data

We use the MAST Disruption Detection Dataset (Sharma 2025, Zenodo 10.5281/zenodo.16032053). The dataset comprises 500 NetCDF shot files, each containing 2000 samples at 4638 Hz spanning ~ 0.43 s of tokamak operation. Ten plasma diagnostics are included per shot: plasma current (I_p), total radiated power, core electron density and temperature, internal inductance, dynamic loop voltage, two D_α channels (HM10, HU10), saddle coil channel 6, and soft-X-ray channel 6. Disruption time, ramp-up boundary, and flat-top boundaries are provided as NetCDF attributes for the 429 disruptive shots; the remaining 71 shots are labelled non-disruptive.

The dataset is not balanced: 429 disruptive vs 71 non-disruptive, roughly 6:1. We mitigate this with class-weight balancing during classifier training, but the relatively small non-disruptive sample is the principal limitation on achievable false-alarm-rate estimation.

4 Iteration log

Table 1: Pareto trajectory across ten iterations. All numbers from 5-fold CV with the same shot indices. TTD = median time-to-disruption. Iteration 9 (**GBM v4 phase, leak-fixed**) is the canonical Pareto champion. Iteration 10 (lean v5) is a documented negative result.

#	Configuration	TPR (%)	FAR (%)	TTD (ms)	Note
1	Unweighted consensus (frac=0.50)	50	25	19	Vanilla baseline
2	Weighted (frac=0.30)	65	77	65	High-recall corner
2	Weighted (frac=0.50)	50	25	12	Equivalent to (1)
3	Hybrid K=2 of 3 (A,B,C gates)	45	32	16	Negative result
3	Hybrid K=3 of 3	11	3	6	High-precision niche
4	Learned LR (7 single-scale feats)	49	14	40	Real gain over hybrid
5	Multi-scale LR (14 feats, 3 scales)	56–71	18–32	53–76	Big lift
6	GBM v1 (14 feats, 150 trees)	66–81	18–44	71–102	GBM dominates LR
7	GBM v2 (dense sampling, 250 trees)	72–83	18–46	71–100	+6 pp TPR at fixed FA
8	GBM v3 (+per-channel peakz, 24 feats)	72–82	21–45	71–100	Plateau
9	GBM v4 (+phase, 27 feats, leak-fixed)	49–73	6–35	65–88	Pareto champion
10	Lean v5 (9 feats from failure analysis)	48–75	14–39	25–39	Negative result

4.1 Negative result: hybrid pendulum/Kuramoto voting

Inspired by an internal research note proposing that wave-interference networks obey nonlinear-pendulum dynamics with golden-ratio equilibria [?], we attempted to combine three structurally distinct gates into a K-of-3 voting rule:

- **Gate A** (coherence-of-derivative): each channel must have $|z_c| > 3$ and $\text{sign}(\dot{z}_c)$ agreeing with the majority of other above-threshold channels.
- **Gate B** (Kuramoto): the order parameter $R(t)$ exceeds its rolling baseline by a configurable ΔR .
- **Gate C** (Hamiltonian rate): $H(t) = \sum_c w_c z_c^2(t)$ has a robust-z-scored $|\dot{H}|$ exceeding threshold.

The K=2 vote rule yielded TPR 45% at FAR 32%, *worse* than the weighted single-gate baseline (45/20) at the same TPR. K=3 carved out a high-precision niche (11/3, precision 96%) but missed too many disruptions to serve as a primary detector. Section 4.2 explains why: on real plasma data the three gates turned out to be correlated, not independent — they all detect “coordinated multi-channel motion” with different math, and non-disruptive events with coordinated motion (ELMs, mode-locking, controlled ramp-downs) fire all three gates simultaneously.

4.2 Recovery via learned-feature classifier

The hybrid failure motivated using the gate scalars as *input features* to a learned classifier rather than voting on them directly. The single-scale 7-feature logistic regression (iteration 4) recovered the loss and added a measurable gain over the weighted baseline: 49% TPR at 14% FAR with 40 ms TTD, a Pareto improvement on all three metrics simultaneously.

The learned coefficients revealed why the K-of-3 vote failed: gate A’s `coherence` coefficient was 0.04, gate C’s `Hamiltonian` coefficient was 0.00, and `max_channel_z` was 0.00. Only `R_excess` (+1.10), `R` (1.25, negative because absolute R is high in both classes), `weighted_consensus` (+1.05) and `n_channels_novel` (0.57) carried weight. The hybrid had attempted to vote on signals that contained no orthogonal information.

4.3 Multi-scale lift

Adding the same features at three baseline scales (50, 200, 500 ms) + two time-derivatives lifted the LR Pareto: TPR 71% at FAR 32% with 76 ms TTD at $p_{\text{thr}} = 0.85$. The 200 ms scale of `weighted_consensus` dominated (LR coefficient +2.03 in the 3-fold average), with 500 ms scales of `R` and `R_excess` contributing the next-largest coefficients. Time-derivatives carried essentially zero weight (+0.05 and +0.01).

4.4 Gradient boosting takes over

A gradient-boosting classifier on the same 14 multi-scale features (iteration 6) further lifted the Pareto: TPR 81% at FAR 44% with 102 ms TTD at $p_{\text{thr}} = 0.85$. Dense training-window sampling (2 ms stride, 50 ms precursor lead, iteration 7) added another ~ 2 pp TPR.

4.5 Phase indicators and a leakage fix

Iteration 9 added three pulse-phase one-hot indicators (`in_rampup`, `in_flattop`, `in_rampdown`). On the first attempt, the model achieved TPR 95% at FAR 92% — a classic feature-leakage signature. The model had discovered that non-disruptive shots are not sampled past their flat-top end in the training sampler, so `in_rampdown` effectively became a perfect label proxy for disruptive vs non-disruptive in the windowed sense.

The fix was to extend non-disruptive shot sampling beyond the flat-top end through the full data range. With the leakage closed, GBM v4 produced the canonical Pareto champion: TPR 73% at FAR 17% with 78 ms TTD at $p_{\text{thr}} = 0.85$ on the seed-42 fold split.

5 Stability cross-validation

We re-ran the GBM v4 5-fold CV with three different random seeds for the fold split (seeds 42, 7, 2024) to confirm the result is robust to which shots happen to land in which fold. Table 2 summarises the per-seed and aggregate statistics.

Table 2: GBM v4 stability across three CV-fold-split seeds. All values in per-cent (TPR, FAR) or milliseconds (TTD). The $p=0.85$ operating point is the canonical Pareto champion; standard deviations are computed across the three seeds.

p_{thr}	TPR mean $\pm$ std	FAR mean $\pm$ std	TTD mean $\pm$ std (ms)
0.85	74.3 $\pm$ 1.0	14.1 $\pm$ 2.3	80.8 $\pm$ 2.4
0.90	66.0 $\pm$ 0.8	7.5 $\pm$ 3.7	71.4 $\pm$ 1.3
0.95	49.3 $\pm$ 0.7	3.8 $\pm$ 1.3	64.2 $\pm$ 0.5

TPR standard deviations are below 1 pp across seeds; TTD standard deviations are below 3 ms. The Pareto champion at $p=0.85$ is **74.3 $\pm$ 1.0% TPR / 14.1 $\pm$ 2.3% FAR / 80.8 $\pm$ 2.4 ms TTD**. This is not a fold-split lottery winner.

6 Failure analysis

We characterise the 24% of disruptions missed at the canonical $p_{\text{thr}} = 0.85$ operating point. For each shot we compute peak feature values in the analysis window and compare disruptive-shot and non-disruptive-shot distributions.

6.1 Discriminating features

Table 3 shows median and quartile statistics for the principal features. The result is sharp:

Table 3: Per-shot peak feature distributions. Disruptive-class median versus non-disruptive-class median. Strong discrimination means a $> 2\times$ ratio in the same direction. *Wrong-direction* features fire *after* disruption, so the windowed pre-disruption pre-image shows the reverse pattern.

Feature	disr. median	non-disr. median	Discrimination
max_weighted_consensus_200ms	0.677	0.266	strong ($2.5\times$)
max_n_novel_500ms (p75)	4.0	1.0	strong upper-tail
max_R_500ms	0.800	0.800	<i>none</i>
max_R_excess_500ms	0.800	0.800	<i>none</i>
saddle_coil_peakz	0.000	0.180	<i>wrong direction</i>
soft_xray_peakz	0.000	0.598	<i>wrong direction</i>
ip_peakz	0.000	0.569	<i>wrong direction</i>

The wrong-direction features are the most physically meaningful diagnosis. The plasma current I_p , the saddle-coil locked-mode amplitude, and the soft-X-ray emission ARE the canonical disruption indicators in the plasma-physics literature. However, they deflect *at* the disruption time, not before. The pre-disruption analysis window ends at t_{disrupt} , so disruptive-shot windows are *quieter* on these channels than non-disruptive flat-top windows (which include normal-physics fluctuations). The classifier therefore uses these channels in reverse: “ I_p quiet $\Rightarrow$ disruption imminent.” This is a fragile signal direction and a key reason for the remaining gap to RNN baselines, which presumably learn a richer within-channel shape feature rather than a simple peak-z reduction.

Equally striking: *the Kuramoto order parameter R itself does not discriminate*. Both classes routinely reach $R \approx 0.8$. Only the *rise of R over a rolling baseline* (i.e., R_{excess}) carries information.

6.2 Structurally invisible disruptions

Defining a disruption as “structurally invisible” if all its peak feature values lie below the 90th percentile of non-disruptive shots, we find $8/417 = 2\%$ of disruptions in this regime. The remaining miss budget at $p_{\text{thr}} = 0.85$ is therefore a threshold-tuning problem, not a missing-information problem.

6.3 Negative experiment: the lean feature set

The failure analysis motivated a 9-feature lean redesign: drop the no-discrimination R -absolute features and the wrong-direction peakz features, add **weighted_consensus** dynamics (rate of rise, duration above 0.3, integral over 500 ms).

Result: TPR 48% at FAR 14% with 25 ms TTD at $p_{\text{thr}} = 0.90$ — much worse than GBM v4 (74/14/81 at $p_{\text{thr}} = 0.85$). The “noisy” features the failure analysis had identified as candidates for removal apparently carried *contextual* information the GBM used effectively. Stripping them sacrificed the model’s ability to disambiguate in specific pulse-phase contexts.

The lesson is that GBM feature-importance rankings are not a clean guide to which features can be discarded: a feature with low marginal importance may still be load-bearing within a specific tree split. Iteration 10 stands as documented evidence of this.

7 Comparison with published baselines

The Ferreira et al. 2019 [?] LSTM on JET bolometer data reports approximately 85% TPR at 15% FAR. Our final operating point is 74.3% TPR at 14.1% FAR — approximately matching the RNN false-alarm-rate while running approximately 11 pp below the RNN’s TPR.

Three caveats apply to this comparison:

1. Different machine (MAST versus JET). MAST disruptions are dominated by density-limit and radiation-collapse precursors; JET disruptions also include locked-mode and tearing-mode causes. Per-machine precursor distributions differ.
2. Different dataset size (500 versus ~ 1500 shots in Ferreira’s evaluation set).
3. Different baseline pipelines. Ferreira uses raw bolometer profiles fed to an LSTM, learning temporal shape implicitly; we use hand-crafted scalar features and gradient boosting, with all temporal structure collapsed to peak/rate/integral summaries.

Closing the remaining 11 pp TPR gap most likely requires one or more of:

- **Operational metadata for multi-class precursor learning.** Disruption-cause labels (radiation, density limit, locked mode, H/L back-transition, ...) would let a multi-class model learn cause-specific precursor signatures rather than a single binary-disruption signature. The current MAST dataset does not expose these labels.
- **Larger training set.** 429 disruptive shots is small for the diversity of physics involved.
- **Sequence-aware architectures.** An LSTM or transformer could learn within-channel *shape* features that our per-window scalar features collapse out. The wrong-direction peakz features identified in §6 would be unaffected because the network would access the time series before disruption directly.

8 Reproducibility

All code, dataset URLs, per-iteration results JSON, and commit history are available at <https://github.com/Zynerji/JET-Disruption>. The canonical Pareto-champion result is reproducible from a clean Python 3.12 environment with `pip install numpy scipy scikit-learn xarray netCDF4` and:

```
git clone https://github.com/Zynerji/JET-Disruption
cd JET-Disruption
python -m pip install netCDF4 xarray scikit-learn
python -c "import urllib.request; urllib.request.urlretrieve(
    'https://zenodo.org/api/records/16032053/files/T_lead_0ms.tar.xz/content',
    'data/T_lead_0ms.tar.xz')"
mkdir -p data && tar -xJf data/T_lead_0ms.tar.xz -C data
python examples/benchmark_mast_gbm_v4.py
python examples/stability_check_v4.py
```

Each benchmark writes a results JSON next to itself. The full Pareto trajectory in Table 1 is the literal output of running iterations 1 through 10 in sequence.

9 Conclusion

We have demonstrated that the Systrophē address-space cross-observable consensus catcher, originally developed for gravitational-wave search and for spacetime-metric special-locus identification, applies to tokamak disruption prediction on real MAST plasma data with measurable performance. The final operating point of $74.3 \pm 1.0\%$ TPR at $14.1 \pm 2.3\%$ FAR with 80.8 ± 2.4 ms median time-to-disruption is within 3 pp of the published RNN false-alarm rate, ~ 11 pp below the RNN true-positive rate, and is statistically stable across fold-split seeds.

The work is documented as an “honest iteration log” with two outright negative results (hybrid voting; failure-analysis-informed feature reduction). The negative results were as informative as the positive ones: they constrained subsequent design choices and exposed a systematic feature-leakage failure mode in the pulse-phase indicator that would have invalidated naive bench-marking.

The remaining gap to RNN baselines is, we argue, in operational-metadata access and sequence-aware shape learning rather than in the consensus methodology itself. The methodology generalises cleanly across domains — gravitational waves, spacetime physics, plasma disruptions — which we take as evidence that cross-observable consensus is a fundamentally good detector design for regime-change problems.

References

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